Problem FoL.1 ... thoughtful

While sitting and thinking deeply, Aleš was throwing a little ball against a wall. Each time, h
hrew the ball with an initial speed $v_0 = 10\,\mathrm{m{\cdot}s^{-1}}$ *at an angle* $\alpha = 60^\circ$ *from the vertical axis*
Since he was sitting in a narrow corridor with width $d = 1.5\,\mathrm{m}$*, the ball always bounced bac*
nd forth between the walls a couple of times. What is the maximum height (measured relativ
o ball's initial position) that the ball will reach along its trajectory? Consider all collisions t
e perfectly elastic. The acceleration due to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$*.*

Mirek's mechanical mental image of periodic boundary conditions

Problem FoL.2 ... Ded Moroz

Consider Mr Spherical (whose shape, incidentally, happens to be perfectly spherical) that like
to wear a tight-fitting jacket of negligible thickness (compared to his radius). Wearing his jacket
Mr Spherical went outside for a walk. Once the thermal equilibrium between Mr Spherica
and his surroundings was established, the temperatures at the inner and outer surfaces of hi
jacket could be measured to be $25\,^{\circ}\mathrm{C}$ *and* $-5\,^{\circ}\mathrm{C}$*, respectively. Find the temperature (in degree*
Celsius), which is measured exactly in the middle of the layer of Mr Spherical's jacket (i.e. hal
way through the bulk of the layer). *Sometimes, Lubošek feels really cold*

Problem FoL.3 ... fiat nox

Consider a short-circuited two-conductor copper wire. At one end of the wire, a resistanc
of $R = 65.3\,\mathrm{m}\Omega$ was measured using an ohmmeter. What is the distance (along the wire) t
the short-circuited section of the wire from the end where the measurement was performed

The resistivity of copper is $\varrho = 1.71 \cdot 10^{-8}\,\Omega\cdot\mathrm{m}$, *the cross-sectional diameter of one conducto*
s $d = 1.50\,\mathrm{mm}$. *A switch was tripped in Mirek's room*

Problem FoL.4 ... the runnier, the better

Dominika is secretly preparing homeopathics at home. She uses a cup filled with a solutio of volume $V_0 = 200\,\mathrm{ml}$ and concentration of the active agent $c_0 = 0.001\,\mathrm{M}$. She then pour away a half of this volume and fills the cup with water to the original volume V_0. She repeat this process 70 times. Statistically speaking, how many molecules of the active agent are lef in the cup at the end? Assume that the solution and water are always perfectly mixed.

Mirek heard some gossip

Problem FoL.5 ... brake failure

A truck experiences brake failure and needs to make an emergency stop. Luckily enough, it'
ust driving through a flood zone with lots of sand-filled bags. Assume that the mass of th
ruck is $M = 6\,\mathrm{t}$ *and that one bag filled with sand weighs* $m = 60\,\mathrm{kg}$*. How many sand bags wi*
he truck need to collide with until it slows down to $75\,\%$ *of its initial speed* $v_0 = 50\,\mathrm{km{\cdot}h^{-1}}$
Assume that the collisions are perfectly inelastic and that after colliding with the truck, th
bags are pushed in front of it without friction.

Mirek was forced to replace people with sand bags

Problem FoL.6 ... fungus librariensis

In order to protect books from becoming damp, some libraries choose to keep the relativ humidity at $\varphi_0 = 25\,\%$. *Find the mass of water which would (after complete evaporation turn such a library into a suitable environment for mould growth. You can assume that th ideal value of relative humidity for mould growth is* $\varphi_p = 60\,\%$. *The total volume of air i this particular library is* $V = 180\,\mathrm{m}^3$, *the (constant) ambient temperature is* $t = 24\,^\circ\mathrm{C}$ *and th saturated vapour density at* $24\,^\circ\mathrm{C}$ *is* $21.8\,\mathrm{g\cdot m}^{-3}$.

Lydka likes to daydream about her private library

Problem FoL.7 ... server not found

Consider computer X, which is connected by two optical fibres to two other computers A and B. The total length of these optical fibres together is $l = 3.0\,\mathrm{km}$. A signal from computer A bearing a message that it was sent at the time $1\,413\,753\,899.213\,575\,3\,\mathrm{s}$ (Unix timestamp of the time counter at computer A) reached computer X at the time $1\,413\,753\,899.213\,592\,1$ (Unix timestamp of the time counter at computer X). Similarly, a signal from computer B was received by computer X at time $1\,413\,753\,906.459\,900\,4\,\mathrm{s}$, bearing the message that it was sent at the time $1\,413\,753\,906.459\,892\,5\,\mathrm{s}$. Assume that the time counters of computers A and B are synchronized, but that the time counter of computer X is either gaining or lagging behind. Luckily, this deviation stays constant in time. Also, do not take into account either the effects of relativity or the time to process the signals. Assuming that the speed of propagation of signals along the fibres is $c = 2.0 \cdot 10^8\,\mathrm{m{\cdot}s^{-1}}$, find the length of the fibre between A and X.

Michal was trying to reduce the dimensions of GPS equations

Problem FoL.8 ... Archimedes drowned

The diagram below shows schematically two situations dealing with an aluminium sphere
ylindrical flask with water and a holding apparatus from which the sphere can be suspended
n the two situations, we have that

*) *the holder is placed on top of the scales with the sphere suspended from it. The sphere i*
fully submerged in the flask which is held off the scales,

b) the same as a), except in this case, the flask rests on the scales.
Find the difference $\Delta m = m_b - m_a$ of the readings m_b and m_a shown by the scales in the si
uation b) and a) respectively.

The radius of the (homogeneous) sphere is $r = 1.03\,\mathrm{cm}$, the inner radius of the flask i $R = 3.01\,\mathrm{cm}$. *The height of the water column in the flask with the sphere fully submerge* *n it is $h = 5.10\,\mathrm{cm}$. The densities of water and aluminium are $\varrho = 996\,\mathrm{kg{\cdot}m^{-3}}$ and ϱ_{Al} =* $= 2\,700\,\mathrm{kg{\cdot}m^{-3}}$, *respectively. The mass of an empty flask is $m_n = 124.5\,\mathrm{g}$. Assume that th* *uspension part of the holding apparatus is of negligible mass and volume. The rest of th* *pparatus has a mass of $m_d = 523.5\,\mathrm{g}$. The range of values that the scales can measure i* *ufficient for it to show both readings with accuracy up to $0.1\,\mathrm{g}$. The density of air should b* *egarded as negligible. State your answer in kilograms.*

Karel wanted to set a simple-minded but computationally horrendous problem

Problem FoL.9 ... carriage on a blind track

Consider a railway carriage with mass $M = 20\,\mathrm{t}$, which moves on the tracks without friction. Inside the carriage, there is an elephant with mass $m = 4\,\mathrm{t}$, initially at rest. Find the distance by which the carriage moves, if the elephant walks a distance of $l = 12\,\mathrm{m}$ and then stops again. (The elephant can move only parallel to the tracks, because the carriage is quite narrow.)

Michal thinks that elephants are cool

Problem FoL.10 ... driving in rain

Find the total volume of water (in litres) that falls down on a Karosa 700 *type bus as it travels fo*
n hour at a speed of $v_b = 50\,\mathrm{km{\cdot}h^{-1}}$. *Assume the bus to be of a cuboid shape with height* $h =$
$= 4\,\mathrm{m}$, *width* $w = 2.5\,\mathrm{m}$ *and length* $l = 12\,\mathrm{m}$. *The rain falls vertically downwards with speed* $v_r =$
$= 10\,\mathrm{m{\cdot}s^{-1}}$ *and flux (that is, the volume of water crossing a unit area perpendicular to th*
elocity of the falling rain, per unit time) $Q = 5\,\mathrm{mm{\cdot}h^{-1}}$. *Xellos running from the the rain*

Problem FoL.11 ... on your own drive

Find the efficiency (in %) of a thermodynamic cycle consisting of two isothermal processes an
wo isochoric processes (see the figure). Assume that the cycle works between temperature
$T_1 = 373\,\mathrm{K}$ *and* $T_2 = 273\,\mathrm{K}$ *and that the heat released during the isochoric process* $2 \to 3$ *i*
ed into the isochoric process $4 \to 1$.

Xellos spied on a hamster rolling his whee

Problem FoL.12 ... the most important letter's centre of mass

Has it ever occurred to you that F *might actually be the most important letter of the alphabet*
Find the y *coordinate of the centre of mass of the shape of the letter* F*. You can assume tha*
he shape is built up from 10 identical homogeneous square-shaped laminae with side length a
Express your answer to three significant figures as a multiple of a *in the coordinate grid indicate*
n the figure. *Karel contemplated on the importance of the letter* F

Problem FoL.13 ... sulphuring the barrels

Sulphur dioxide was and still is used as a preservative for wine casks. One of the sulphurin techniques involves burning sulphur discs directly inside the casks. Consider a cask with volu me 200 l, *inside which two sulphur discs are placed, each containing* 2 g *of sulphur. The disc are then ignited and the cask is closed hermetically. What is the fraction of cask's volum that* SO_2 *will occupy after both discs burn down completely? Assume that when all burnin stops, the temperature inside the cask decreases immediately back to* $T = 20\,°C$, *the same as i was initially. Molar mass of sulphur is* $M = 32\,g{\cdot}mol^{-1}$ *and the ambient atmospheric pressur is* $p = 101\,kPa$. *Mirek knows a thing or two about physicists' views on chemical calculations*

Problem FoL.14 ... to jump, or not to jump

A horizontal plate performs simple harmonic motion with period $T = 0.5\,\mathrm{s}$ in the vertica direction. Find the maximum value of the amplitude of oscillations A (in centimetres) suc that a body lying on the plate would remain in contact with the plate throughout the motion The acceleration due to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$.

f(Aleš) salvaged this from an ancient book

Problem FoL.15 ... tea at five

Náry decided to make himself a cup of mint tea. Reading the instructions „Place the teabag int boiling water.“ on the side of the box made him think about the following question: How muc energy could be saved, if only he could prepare his tea at the summit of Triglav, the highes peak in Slovenia (2 864 meters above the sea level) instead of doing it at his home in Boskovice You can assume that the ambient pressure and temperature in Boskovice are $p_B = 101\,325\,\mathrm{P}$ and $t_B = 25\,^\circ\mathrm{C}$ respectively. The temperature at Triglav is $t_T = 13\,^\circ\mathrm{C}$ and ideal gases ther occupy 1.44 times the volume they occupy in Boskovice. Also, the boiling point of water t_v an the ambient pressure p are related (over a suitable range of values of p) as

$$t_v = \tau + kp\,,$$

where $\tau = 71.6\,^\circ\mathrm{C}$ and $k = 2.8 \cdot 10^{-4}\,\mathrm{K{\cdot}Pa^{-1}}$. In order to prepare his tea, Náry needs t bring $m = 600\,\mathrm{g}$ of water with initial temperature $t_0 = 20\,^\circ\mathrm{C}$ (assumed to be the same i both cases) to a boil. All energy losses should be neglected. The specific heat capacity of wate is $c = 4\,180\,\mathrm{J{\cdot}kg^{-1}{\cdot}K^{-1}}$. *Kiki and Tom pondering about their excessive consumption of te*

Problem FoL.16 ... scary shopping

Kiki set off to London (which lies $s = 1\,500\,\mathrm{km}$ *from her home) to buy some* $^{18}\mathrm{F}$*. Knowing tha the half-life of this poor quality stuff was only* $1.8\,\mathrm{h}$*, she decided to buy* $32\,\mathrm{g}$ *right away. Find th speed (in* $\mathrm{km}\cdot\mathrm{h}^{-1}$*) with which she had to fly back in order to bring home* $10\,\mathrm{g}$ *of* $^{18}\mathrm{F}$ *more tha f she had opted for a 24 hour journey by bus. Dominika had to go and claim her mone back already a third time this year, wondering why anyone else's shoes last so much longe*

Problem FoL.17 ... feed a castle

n their time (i.e. at the turn of the 19th and 20th century), the owners of the Kokořín castl vere progressively thinking people – they introduced gas heating based on burning acetylen reated by reacting calcium carbide with water

$$\mathrm{CaC_2} + 2\,\mathrm{H_2O} \longrightarrow \mathrm{C_2H_2} + \mathrm{Ca(OH)_2}\,.$$

Compare the energetic efficiency of this carbide-fuelled heating to that of a wood-fuelled alter native, i.e. find the ratio of energies gained from a unit mass of wood and a unit mass of carbide Use the following data: density of calcium carbide $\varrho(\mathrm{CaC_2}) = 2\,200\,\mathrm{kg{\cdot}m^{-3}}$, *molar mass of calci um carbide* $M(\mathrm{CaC_2}) = 64\,\mathrm{g{\cdot}mol^{-1}}$, *heating value of acetylene* $H(\mathrm{C_2H_2}) = 48\,\mathrm{MJ{\cdot}kg^{-1}}$, *mola mass of acetylene* $M(\mathrm{C_2H_2}) = 26\,\mathrm{g{\cdot}mol^{-1}}$, *heating value of wood* $H_d = 14\,\mathrm{MJ{\cdot}kg^{-1}}$, *density o wood* $\varrho = 520\,\mathrm{kg{\cdot}m^{-3}}$. *Assume that the castle has a sufficiently large water reservoir. The fac hat the fuel needs to be transported over an altitude difference* $h = 50\,\mathrm{m}$ *from a nearby lyin illage uphill should be included in your energy balance considerations. The acceleration du o gravity is* $g = 9.8\,\mathrm{m{\cdot}s^{-2}}$.

Mirek and Lukáš on a trip

Problem FoL.18 ... bullet

In a dark abandoned hall, there are a robber and a policeman standing against each other b
the opposite sides of the hall. Having heard steps echoing in the darkness, the criminal point

his gun horizontally on the unsuspecting cop and is about to shoot. What he doesn't know s that a potential difference $U = 10\,\mathrm{kV}$ was brought between the floor and the ceiling of th all, thus creating a homogeneous field acting (vertically downwards) on the fired bullet, sinc he bullet acquires a charge q as it travels through the muzzle. Find the least charge-to-mas atio q/m of the bullet such that it hits the floor before it reaches the policeman (where by m ve denote the mass of the bullet). The robber and the policemen are separated by a distanc of $D = 100\,\mathrm{m}$, the ceiling is $d = 8\,\mathrm{m}$ above the floor and the bullet is fired horizontally wit muzzle velocity $v_0 = 400\,\mathrm{m{\cdot}s^{-1}}$. Assume that the robber holds his gun $h = 1.5\,\mathrm{m}$ above th loor and that the acceleration due to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$.

Mirek wanted to point out the importance of triboelectric effec

Problem FoL.19 ... watermelon

Náry was sitting on a swing, which was at rest. Suddenly, he felt like eating a watermelon. Two other FYKOS guys immediately threw him one with mass $m_m = 20\,\mathrm{kg}$. Assuming that the speed of the watermelon was $v_m = 10\,\mathrm{m\cdot s^{-1}}$, find the angle by which the swing with Náry moves. Náry (approximated by a point particle) weighs $m_n = 80\,\mathrm{kg}$ and the swing rope is $L = 4\,\mathrm{m}$ long. Also, assume that the watermelon hit Náry (who managed to catch it) horizontally in the plane of rotation of the swing. State your answer in degrees. Gravitational acceleration is $g = 9.81\,\mathrm{m\cdot s^{-2}}$.

Lydka was eating a watermelon

Problem FoL.20 ... a superconducting one

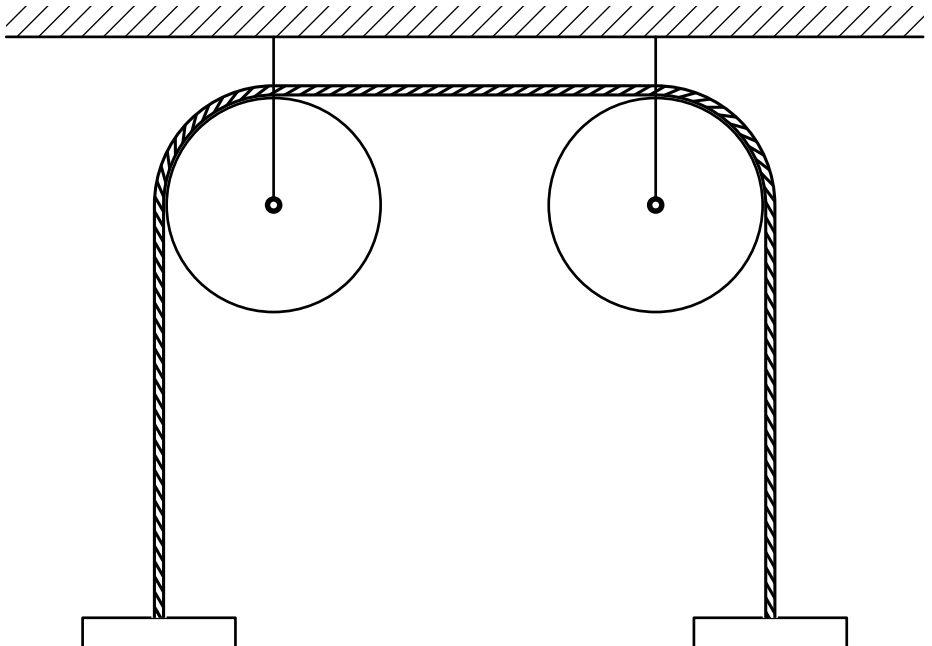
Consider two superconducting capacitors with capacitances $C_1 = 10\,\mu\mathrm{F}$, $C_2 = 5.0\,\mu\mathrm{F}$ *an*
a superconducting DC power supply with voltage $U = 10\,\mathrm{V}$. *All these components are connecte*
by superconducting wires. The superconductors are perfect, hence there is no resistance an
Therefore, Joule heating will not occur. First, let us connect the first capacitor to the DC suppl
and wait until the charge builds up. When it is fully charged, we disconnect the DC power suppl
and connect the second capacitor to the first one instead. Find the difference between the energ
stored in the first capacitor alone and the total energy of the two connected capacitors.

Even the written exams can be educating for Mirek

Problem FoL.21 ... buoyant swings with a pulley

Consider two identical cylinders (base radius 5 cm, *height* 10 cm *and density* $\varrho_0 = 2.2\,\mathrm{g{\cdot}cm^{-3}}$ *in equilibrium, partially submerged in two containers filled with liquids of densities* $\varrho_1 = 1.0\,\mathrm{g{\cdot}cm^{-3}}$ *and* $\varrho_2 = 0.8\,\mathrm{g{\cdot}cm^{-3}}$ *(see the diagram below). Find the period of small vertical oscillations of this system about its equilibrium. Neglect the changes in the level of liquids in the containers. Neglect also the mass of the strings. The acceleration due to gravity is* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$.

Due to Jančí



Problem FoL.22 ... on the head

A new notice board needs a new nail (with length $l = 80\,\mathrm{mm}$), with which it should be naile to the wall using our hammer with mass $m = 0.4\,\mathrm{kg}$. Five hits in total were needed to d this. During each hit, the speed of the hammer just before it hit the head of the nail was $v = 8.0\,\mathrm{m{\cdot}s^{-1}}$ (assume that upon hitting the nail, the hammer transfers all its kinetic energy int the nail). However, having nailed it down, we noticed that the board hangs askew. What i the magnitude of the force, with which we will need to pull the nail out of the wall? Assum that the mass of the nail is negligible. Also, assume that the magnitude of the frictional forc s directly proportional to the length of the part of the nail, which is inside the board, and tha t is the same during both pulling the nail out and nailing it in.

f(Aleš) was cleaning his notice boar

Problem FoL.23 ... an infinite circuit

Find the resistance between the terminals A *and* B *as shown on the diagram below. Put* $R =$ $= 5\,\Omega$, $r = 1.5\,\Omega$ *and neglect the resistance of leads.*

Xellos found solving the infinite resistor ladder too easy

Problem FoL.24 ... bowling with a physicist

Once, when Olda went to play bowling, an intriguing thing happened. He threw the bowlin ball of mass $7.25\,\mathrm{kg}$ *and radius* $0.108\,\mathrm{m}$ *with speed* $12\,\mathrm{m{\cdot}s^{-1}}$ *and gave it such a spin tha the ball stopped and stayed put at rest before reaching the skittles. What was the angula speed (in radians per second) which Olda had to give the ball for this to occur? Gravitationa acceleration is* $9.81\,\mathrm{m{\cdot}s^{-2}}$*. Air resistance and rolling friction can be neglected.*

Organizers went bowling

Problem FoL.25 ... waves and wavelets

As it known, propagation of surface waves on a lake is not significantly influenced by gravity i
he short-wavelength limit. Nevertheless there is a strong effect of surface tension which need
o be considered. Using dimensional analysis, determine the angular frequency of surface wave
s a function of surface tension σ, water density ϱ and the wave number $k = 2\pi/\lambda$, where λ
s the wavelength. Assume that the dimensionless coefficient is equal to one in this case. Also
derive an expression for the group velocity and substitute the values $\sigma = 73\,\mathrm{mN{\cdot}m^{-1}}$, ϱ =
$= 1\,000\,\mathrm{kg{\cdot}m^{-3}}$, $\lambda = 2\,\mathrm{cm}$.

Mirek trying to catch short waves

Problem FoL.26 ... rolling cylinder

As a proper engineer, Dominika often indulges herself with rolling homogeneous cylinders dow
a hill. At one such occasion (having a moment of clarity) she came up with an alternative metho
of determining the number of turns the cylinder completes before it falls off an inclined plane
First, she measured its mass ($m = 123.8\,\mathrm{g}$) and also its moment of inertia $J = 1.807{\cdot}10^{-5}\,\mathrm{kg{\cdot}m^2}$
It is widely known that Dominika's inclined plane is $l = 1.02\,\mathrm{m}$ long and that it makes an arbit
rary angle α with horizontal. How many turns (real number, to three significant figures) doe
the cylinder complete before it reaches the end of the inclined plane? Assume that the cylinde
rolls precisely from one end of the plane to the other and that it moves without slipping in th
direction of maximum slope.

Karel wondered about Dominika's future in the field of engineering

Problem FoL.27 ... Zdeněk's sphere

As a proper soon-to-be-engineer, Zdeněk (like Dominika) likes to roll various objects down an in
ined plane. Nevertheless, unlike Dominika, who is rather more into cylinders, Zdeněk prefer
o play with his sphere. He is not a fan of ordinary experimental methods, so in order to fin
ut the density of his sphere, he decided to measure its mass $m = 123.8\,\mathrm{g}$ *(which is incidentall*
he same as that of Dominika's cylinder) and its moment of inertia $J = 3.48 \cdot 10^{-5}\,\mathrm{kg{\cdot}m^2}$*. You*
ask is to help Zdeněk with his calculation and determine the density of his solid homogeneou
phere.

Karel fantasizing about BUT..

Problem FoL.28 ... energetic surface

What is the contact angle (in degrees) of the interface of air, water (surface tension γ_l = = $74\,\mathrm{mN{\cdot}m^{-1}}$) and an unknown solid with surface energy (sc. the solid-air interfacial ener gy) $\gamma_s = 53\,\mathrm{mN{\cdot}m^{-1}}$? The solid-water interfacial energy is $\gamma_{sl} = 25\,\mathrm{mN{\cdot}m^{-1}}$.

A by-product of Terka's work on a FYKOS experimental task

Problem FoL.29 ... pyramid's circuits vol. 2

n the circuit shown below we first put the switch on and wait for the currents to come t steady state. Then we put the switch back off. What is the magnitude of the potentia lifference (in volts) measured by the voltmeter immediately after the switch was turned off Take the capacitors and coils to be ideal and assume that the voltmeter provides an infinit esistance. Put $\varepsilon = 1\,\mathrm{V}$ *and* $R_1 = 3R_2$.

Xellos found out about the Estonian-Finnish Olympiad in Physics

Problem FoL.30 ... gassed joules

Náry once passed by a pressure vessel containing sulphur dioxide to which a number of meter were attached and some data were printed on its mantle. Namely, he managed to read off tha the volume of the vessel was $V = 1\,\mathrm{m}^3$ and that the mass of an empty vessel would be m = = 100 kg. The meters were showing the temperature of the gas inside, its pressure $p = 5.3\,\mathrm{MP}$ as well as the combined mass of the gas and the vessel $M = 152\,\mathrm{kg}$. However, being myopic, Nár was unable to read off the value of the temperature. Luckily, he had 4 hours of free time so h set off to calculate the temperature assuming there was an ideal gas inside the vessel. Deemin the value he obtained too low and having another 6 hours free, he decided to recalculate th temperature under the assumptions of van der Waals theory. What is the magnitude of th difference of the two results he got (in kelvins)?

f(Aleš) remembered his Thermodynamics example classes

Problem FoL.31 ... elusive lens

During a clean-up day in the FYKOS room, a thin lens together with a circular-aperture ligh
ource with diameter $d = 5\,\mathrm{mm}$ *were found. First consider placing the lens in a distance* s_1 =
$= 12\,\mathrm{cm}$ *from a screen. If we then put the source in a large distance from both the screen an*
he lens, we will observe a point-like image on the screen. If on the other hand we place th
ens and the source in a distance s_2 *and* l_2 *respectively from the screen, we will observe a shar*
ircular image with diameter $d_2 = 2\,\mathrm{cm}$. *Find the distance* l_2 *(in* cm*).*

Outrageously, Xellos prefers doing experiments rather than helping with clean- up

Problem FoL.32 ... playing in the field

Maťko and Kubko, two neon isotopes with mass numbers $A_M = 20$ and $A_K = 22$, came up wit
a new game, where Maťko simply tries to copy all moves of Kubko. Carrying a charge of $+e$
Kubko entered a homogeneous magnetic field with magnitude $B = 0.24\,\mathrm{T}$ in the directio
perpendicular to the field lines. Maťko, who also carries a charge of $+e$, entered the fiel
at the same point and in the same direction as Kubko did. Find the distance d of the points wher
the respective isotopes left the field, provided they both have the same kinetic energy E_K =
$= 6.2 \cdot 10^{-16}\,\mathrm{J}$. *Take the values of the atomic mass unit and the elementary charge to be m_u =*
$= 1.660\,57 \cdot 10^{-27}\,\mathrm{kg}$ *and* $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$. *Lydka likes to play with fridge magnets*

Problem FoL.33 ... planet Fykosia

Fykosia is an as yet undiscovered planet of our Solar System. Find the ratio a/r (where r i the radius of Fykosia and a is its distance from the Sun) and submit your answer as a multipl of π. Assume that the trajectories along which the planets orbit the Sun are circles and that th Sun lies in their common centre. The orbital period of Fykosia is $T = 3\sqrt{3}\,\mathrm{yr}$ and its densit is $\varrho = 5\,000\,\mathrm{kg{\cdot}m^{-3}}$. Also, assume that the magnitude of the force due to gravity acting on body with mass $m = 1\,\mathrm{kg}$ placed on the surface of Fykosia is $F_G = 69\,\mathrm{N}$. The gravitationa constant is $G = 6.67 \cdot 10^{-11}\,\mathrm{N{\cdot}m^2{\cdot}kg^{-2}}$.

Fykosie, my homeland

Problem FoL.34 ... feather-light

Imagine yourself in the role of an observer situated in the direction of north pole of a galax called Fykopaedia, which you observe rotating clockwise. Once you have managed to identif what appears to be the galactic centre, you also spot a star orbiting the centre along a circula trajectory with radius $r = 7\,500\,\mathrm{pc}$ at the very edge of the galaxy. Having obtained a spectrum of this star, you notice that a hydrogen line with laboratory wavelength $\lambda_{emit} = 587.49\,\mathrm{nm}$ i transverse Doppler-shifted to $\lambda_{obs} = 589.89\,\mathrm{nm}$. Your task is now to find the mass of Fykopaedi in the units of solar mass. You should disregard any effects due to possible presence of dar matter or dark energy. Also, assume that the observed spectral shift is solely due to transvers Doppler effect and that the mass distribution in the galaxy is spherically symmetric.

f(Aleš) complaining about his heavy bag

Problem FoL.35 ... an accelerated one

Consider a proton travelling inside a synchrotron with a total energy of $E_p = 1.00\,\mathrm{GeV}$. *The pro*
on collides with an alpha particle with a total energy of $E_\alpha = 3.75\,\mathrm{GeV}$. *Given the rest masse*
of a proton and an alpha particle $m_p = 0.938\,\mathrm{GeV}/c^2$ *and* $m_\alpha = 3.727\,\mathrm{GeV}/c^2$, *respectively*
determine the ratio v_p/v_α *of the speeds of the particles before they collide.*

Mirek likes getting results which resemble fundamental mathematical constants

Problem FoL.36 ... resistance upside down

We want to build a circuit with total resistance $R = (57\,688/1\,807)\,\Omega$, but we only have resistor with resistance $r = 1\,\Omega$ at our disposal. On top of that, we want to build our circuit so tha in each step (starting with one resistor) we connect only one resistor to an already existing par of the circuit, either in series or in parallel. What is the smallest number of resistors we nee to reach our goal?

Xellos recalled a physically motivated problem from a programming competition

Problem FoL.37 ... nomen omen

During a meeting of the greatest villains of all time, The Emperor was laughed at: his Deat Star has a word "star" in its name, yet it does not shine. He could not possibly let suc insolence pass and decided to fix the problem. His Death Star now emits light as if it wa a gray body with emissivity $\varepsilon = 0.8$ *across all wavelengths. (Emissivity of a body is define to be the ratio of intensity of radiation emitted by the body to the intensity of radiatio emitted by a black body at the same temperature.) The radius of the Death Star is* $r =$ $= 450\,\mathrm{km}$ *and an inspecting spaceship, being at a distance of* $d = 10\,000\,\mathrm{km}$ *from the Star measured intensity* $I = 33\,\mathrm{kW{\cdot}m^{-2}}$ *of radiation coming from the Star. What is the wavelengt (in nanometres) at which the Death Star radiates most of its energy? Stefan-Boltzmann constan is* $\sigma = 5.67 \cdot 10^{-8}\,\mathrm{W{\cdot}m^{-2}{\cdot}K^{-4}}$*, Wien's displacement constant is* $b = 2.90 \cdot 10^{-3}\,\mathrm{m{\cdot}K}$.

Mirek lives by the philosophy that there is never enough Star Wars themed problems

Problem FoL.38 ... pendulum

Once upon a time there lived an infinite superconducting horizontal plane with surface mas
density $\sigma = 2 \cdot 10^{10}\,\mathrm{kg{\cdot}m^{-2}}$ *and a pivot placed* $d = 5\,\mathrm{m}$ *above the plane. And so it came t*
pass that somebody came up with the idea to suspend an electron on a string with length $l =$
$= 1\,\mathrm{m}$ *from the pivot. Find the period of small oscillations (in seconds) of the electron abou*
ts equilibrium.

Assume that charge is free to move within the plane arbitrarily fast. Neglect any effects du
o magnetic field.

Xellos walking an electron on a dog lead

Problem FoL.39 ... magnetic focusing

Assume that we fire a beam of protons from a given location on the x axis. In the beam every proton has a speed of $v = 10^5\,\mathrm{m{\cdot}s^{-1}}$, whose direction deviates from that of the x axi by at most $\Delta\alpha = 1°$. Assume that there is an electric field pointing in the x direction wit magnitude $E = 12\,\mathrm{N{\cdot}C^{-1}}$, together with a magnetic field with magnitude $B = 1\,\mathrm{mT}$, als pointing in the x direction. The protons will first come back to the x axis at a distance fro the firing point which lies in an interval of the form $(L, L + \Delta L)$. Find the ratio $(\Delta L)/L$. D not take into account the protons fired precisely along the x axis. *Xellos was playing Bang*

Problem FoL.40 ... slingshooty

The Czech mythical genius Jára Cimrman is fabled for his construction of the so called Čakovic
slingshot. Being curious, we loaded the slingshot with a projectile and fired. Having grossl
underestimated its power, not only did we find that the projectile did not land in Čakovic
(now a district of Prague), we discovered that it basically did not land at all. It went throug

he Earth's atmosphere, across the Solar system, all the way out of the Galaxy, even escapin he local group, passing through the Virgo cluster... until it reached the very borders of th isible universe and ended up in an alternative one.

The physical laws of this alternative universe are quite similar to those of our universe, excep hat the gravitational force between the projectile and rest of the matter in this alternativ niverse, while still derived from the Newton's law, acts repulsively. Consider the projectil pproaching one of local stars. If there was no interaction between the projectile and the sta he projectile would miss the star at distance $b = 10^{10}\,\mathrm{m}$. *However, as a result of the abov lescribed interaction, the trajectory of the projectile will be deflected. Assuming that the mas f the projectile is* $m = 1\,\mathrm{kg}$, *that the mass of the star is* $M = 10^{30}\,\mathrm{kg}$ *and that the spee f the projectile far away from the star is* $v_0 = 10^5\,\mathrm{m{\cdot}s^{-1}}$, *find the minimum separation o he projectile and the star (both of which are to be regarded as point particles) during thei nteraction. The gravitational constant is* $G = 6.67 \cdot 10^{-11}\,\mathrm{m^3{\cdot}kg^{-1}{\cdot}s^{-2}}$.

Mirek applied Rutherford's experiment in the macroworld

Problem FoL.41 ... little gifts

Having left the student housing, Mirek's roommate left behind a number of interesting objects amongst them a 10 m *long rope,* 170 cm *long iron rod and a bag full of cobble stones. Now, let's consider the following situation: first, let's tie the rope to one end of the rod and fix the bag with cobble stones to the same end. Then throw the rod with the bag out of the window, while holding the other end of the rope firmly. The rod lands on the pavement (which is* $d = 11$ m *below the window) in such a way that its free end rests on the pavement exactly below the point where the other end of the rope is held. Assuming that the rod can rotate freely about the end on which it rests, but that the end itself cannot move (there is an infinite friction between the pavement and the rod), find the tension in the rope in equilibrium. The bag with cobble stones has mass* $m_k = 8\,\mathrm{kg}$*, the rod weighs* $m_t = 5\,\mathrm{kg}$ *and the stiffness of the rope is* $k = 200\,\mathrm{kg{\cdot}s^{-2}}$*. Neglect the mass of the rope.*

And don't forget to fix the road afterwards! (Mirek)

Problem FoL.42 ... van de Graaff

Consider a conducting sphere with radius $R = 7\,\mathrm{cm}$ *charged to a potential* $\varphi = -5\,\mathrm{V}$ *(take*
relative to a point at infinity) and covered with a thin layer of dielectric. The sphere is place
into a chamber with ionised helium He^+ *where we assume that only the ions with negligibl*
kinetic energy stick to the surface of the sphere. Finally, the sphere is moved into an evacuate
chamber with volume $V = 10\,\mathrm{m}^3$ *and quickly discharged. Find the equilibrium temperatur*
(in kelvins) of the ion gas thus created in the chamber. *Xellos loves electroshocks*

Problem M.1 ... electrical shopping cart

Being incredibly lazy, the organisers of FYKOS decided to construct a small self-propelled carg vehicle. It remains to fine tune the engine. Having a total mass of $m = 70\,\mathrm{kg}$*, the vehicle i known to accelerate with* $a = 1\,\mathrm{m{\cdot}s^{-2}}$ *when it moves horizontally. Find the vehicle's (uniform acceleration on a* 10% *grade slope assuming that the same thrust is applied in both cases. Th acceleration due to gravity is* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$*.* *f(Aleš) transporting the competition prizes*

Problem M.2 ... Náry's academic quarter

Most people's reactions can be virtually immediate but not in the case of Náry. Even when i comes to mere releasing of things to fall freely under gravity, it takes him $\tau = 6\,\mathrm{s}$ to react o signal. Find the time (in seconds) which elapses until the distance between a body release by Náry and a body released immediately on signal grows to $l = 200\,\mathrm{m}$. The acceleration du to gravity is $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$ and we start measuring time at the exact moment when the firs body is released.

f(Aleš) lamenting Náry's late arrivals. .

Problem M.3 ... truly solid lecture notes

As we all know, some books do not really count as light-weights, so instead of carrying them in a bag let's try and drag them on the floor on a string. Consider a book lying on the floor and a string with length $l = 2.0\,\mathrm{m}$ *whose one end is attached to the centre of the upper side of book's cover. The other end of the string is held in height* $h = 1.2\,\mathrm{m}$ *above the floor. Find the coefficient of friction between the book and the floor knowing that the book which is of mass* $m = 1\,\mathrm{kg}$ *is dragged at constant speed with force of magnitude* $F = 5\,\mathrm{N}$*. Assume that the book does not open. The acceleration due to gravity is* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$*.*

f(Aleš)'s bag is getting heavier and heavier

Problem M.4 ... all in vain

Consider a long pipe whose one end is plunged into water. Inside the pipe, there is a piston which perfectly fits between the walls. Having squeezed the air out, we start pulling the piston up, thus lifting a column of water in the pipe under the piston head. What is the maximum height h in which we can lift the water assuming that the ambient atmospheric pressure is p_a = = 760 mmHg?

f(Aleš) was short of water

Problem E.1 ... the long one and the wide one

Consider two homogeneous cylindrical copper wires of equal mass, one of them being thric
as long as the other. Find the ratio of electrical resistances along the two wires, longer to th
shorter.

Lukáš likes to steal lab equipment

Problem E.2 ... one, two, three,...

Náry was asked to find out the number of turns in his brand new copper coil. Too lazy t count them one by one, he resorted to his multimeter and good old data tables. Having brough a potential difference $U = 7.8\,\mathrm{V}$ across the coil, he measured a current $I = 2\,\mathrm{A}$ flowing in th coil. Also, he measured the cross-sectional diameter of the wire to be $S = 1\,\mathrm{mm}^2$. He the used his data tables to find the electrical resistance of copper to be $\varrho = 1.8 \cdot 10^{-7}\,\Omega\cdot\mathrm{m}$. He als measured the diameter of the turns and obtained an average value $d = 6\,\mathrm{cm}$. Assuming tha the total length of leads is $l_0 = 18\,\mathrm{cm}$, find the number of turns in Náry's coil.

f(Aleš) wanted to set a black-box problem

Problem E.3 ... experimental afternoon

The power outlets in a building where a FYKOS camp was held provide voltage $U_0 = 230\,\mathrm{V}$ *However, some of the experiments had to be performed outside, especially those which involve shooting with an air gun. To this end, an extension lead of length* $l = 101\,\mathrm{m}$ *and resistance* $R_0 = 1.4\,\Omega$ *was used to bring the power supply where it was needed. Consider connecting a load to the extension lead. The properties of the load are such that if we bring a potential difference* $U_s = U_0$ *to it, the load provides power* $P_s = 1\,000\,\mathrm{W}$ *to an appliance (e.g. a heating spiral). Assuming that voltage* $U_s = U_0$ *is brought to the other end of the lead, find the potential drop across the load.*

f(Aleš) had a moment of nostalgia

Problem E.4 ... pyramid's circuits

n the circuit shown below we first put the switch on and wait for the currents to come t steady state. Find the magnitude of the potential difference (in volts) measured by th oltmeter. Take the capacitors and coils to be ideal and assume that the voltmeter provides a nfinite resistance. Put $\varepsilon = 1\,\mathrm{V}$. The resistance $R_1 = 3R_2$.

Xellos found out about the Estonian-Finnish Olympiad in Physics

Problem X.1 ... uncertain bacteria

Use a quantum uncertainty principle to find the uncertainty Δv_x *in the component* v_x *of th velocity of a bacterium with mass* $m = 1.0 \cdot 10^{-15}\,\mathrm{kg}$ *and linear size* $l = 1.0 \cdot 10^{-6}\,\mathrm{m}$. *Assume tha the position of the bacterium can be determined with uncertainty* $\Delta x = l/100$ *and that* $v_x =$ $= 3.0 \cdot 10^{-5}\,\mathrm{m{\cdot}s^{-1}}$. *Submit your answer as* $\log_{10}(\Delta v_x / v_x)$.

Mirek was trying to figure out why it is so difficult to determine the origin of some diseases

Problem X.2 ... a lecture on relativity

Two professors pass each other in a corridor at the Institute of Theoretical Physics, going i
pposite directions. One is coming back from a lecture about special theory of relativity an
is speed is $v_s = 1\,\mathrm{m{\cdot}s^{-1}}$*, while the second one is going to a lecture about general relativit*
nd his speed is $v_g = 2\,\mathrm{m{\cdot}s^{-1}}$*. Determine the error that we will make by using classical physic*
nstead of relativity to calculate the speed of the second professor in the reference frame of th
irst professor. *Mirek trying to entertain himself while waiting for a lecture on relativity*

Problem X.3 ... hot grating

Determine the temperature needed to observe diffraction of a mono-energetic $^{20}\mathrm{Ne}$ *beam (pro duced by an energy filter) on a grating with slit spacing* $d = 0.5\,\mathrm{nm}$*. The condition for diffractio to be observed is taken to be* $\lambda = d$ *where* λ *is the de Broglie wavelength of the atoms in th beam.*

Mirek skimming through a quantum physics textbook

Problem X.4 ... beta decay

Consider a beam of neutrons, where every neutron in the beam has kinetic energy $E_k = 0.05\,\mathrm{eV}$ *Take the half-life and the rest energy of a neutron to be* $T_{1/2} = 640\,\mathrm{s}$ *and* $m_n c^2 = 940\,\mathrm{MeV}$ *respectively. Find the fraction of the number of neutrons which will decay before the beam travels a distance of* $d = 10\,\mathrm{m}$. *Mirek got inspired by a textbook about particle physics*